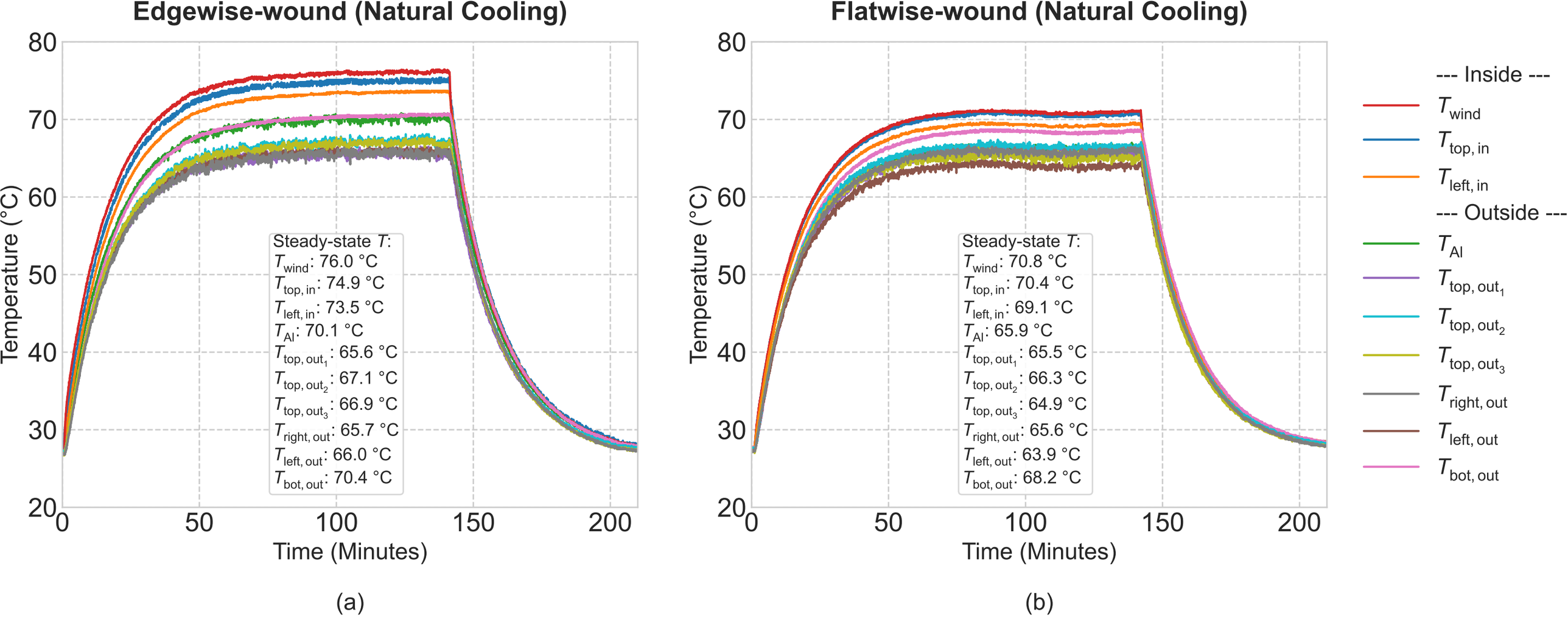
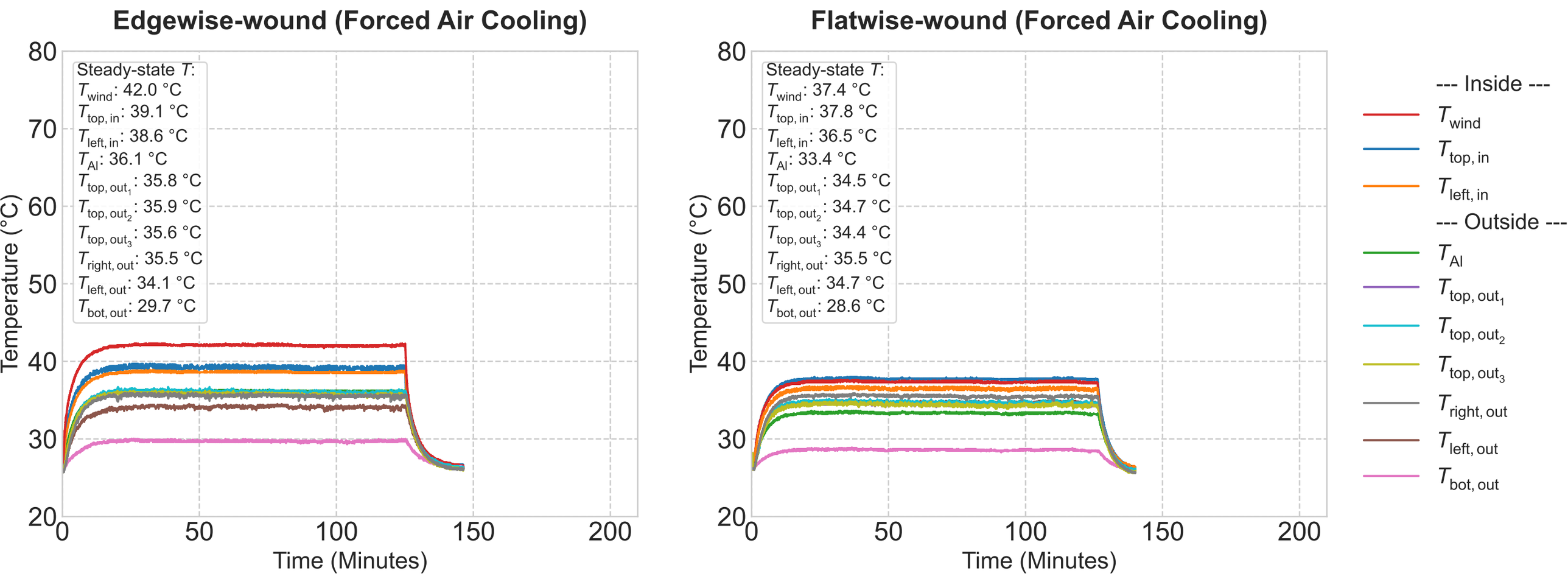

Edgewise-wound (Natural Cooling)
Temperature (°C)
Time (Minutes)
Steady-state T:
T_wind: 76.0 °C
T_top, in: 74.9 °C
T_left, in: 73.5 °C
T_Al: 70.1 °C
T_top, out1: 65.6 °C
T_top, out2: 67.1 °C
T_top, out3: 66.9 °C
T_right, out: 65.7 °C
T_left, out: 66.0 °C
T_bot, out: 70.4 °C
Flatwise-wound (Natural Cooling)
Steady-state T:
T_wind: 70.8 °C
T_top, in: 70.4 °C
T_left, in: 69.1 °C
T_Al: 65.9 °C
T_top, out1: 65.5 °C
T_top, out2: 66.3 °C
T_top, out3: 64.9 °C
T_right, out: 65.6 °C
T_left, out: 63.9 °C
T_bot, out: 68.2 °C
--- Inside ---
T_wind
T_top, in
T_left, in
--- Outside ---
T_Al
T_top, out1
T_top, out2
T_top, out3
T_right, out
T_left, out
T_bot, out

(a) (b)


Edgewise-wound (Forced Air Cooling)
Temperature (°C)
Time (Minutes)
Steady-state T:
T_wind: 42.0 °C
T_top, in: 39.1 °C
T_left, in: 38.6 °C
T_Al: 36.1 °C
T_top, out1: 35.8 °C
T_top, out2: 35.9 °C
T_top, out3: 35.6 °C
T_right, out: 35.5 °C
T_left, out: 34.1 °C
T_bot, out: 29.7 °C
Flatwise-wound (Forced Air Cooling)
Steady-state T:
T_wind: 37.4 °C
T_top, in: 37.8 °C
T_left, in: 36.5 °C
T_Al: 33.4 °C
T_top, out1: 34.5 °C
T_top, out2: 34.7 °C
T_top, out3: 34.4 °C
T_right, out: 35.5 °C
T_left, out: 34.7 °C
T_bot, out: 28.6 °C
--- Inside ---
T_wind
T_top, in
T_left, in
--- Outside ---
T_Al
T_top, out1
T_top, out2
T_top, out3
T_right, out
T_left, out
T_bot, out

(c) (d)